

Linear Programming and its Applications

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Topic: Word problem on Simplex method.

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Que. A farmer owns a 100-acre farm and plans to plant at most three crops. The seed for crops A, B and C costs \$40, \$20 and \$30 per acre, respectively. A maximum of \$3200 can be spent on seed. Crops A, B and C require one, two and one work days per acre, respectively, and there are a maximum of 160 work days available. If the farmer can make a profit \$100 per acre on crop A, \$300 per acre on crop B, and \$200 per acre on crop C, how many acres of each crop should be planted to maximize profit?

Soln:- Total available land = 100 acres.

Crop	Seed cost per acre (\$)	Work days per acre	Profit per acre (\$)
A	40	1	100
B	20	2	300
C	30	1	200

Total available seed budget = \$ 3,200

Total available work days = 160

Formulation:

Let x_1 be the number of acres of crop A.

x_2 be the number of acres of crop B.

x_3 be the number of acres of crop C.

$$\begin{aligned}
 \text{Max } Z &= 100x_1 + 300x_2 + 200x_3 \\
 \text{s.t.} \quad &40x_1 + 20x_2 + 30x_3 \leq 3200 \\
 &\quad \text{(Seed cost constraint)} \\
 &x_1 + 2x_2 + x_3 \leq 160 \\
 &\quad \text{(Work days constraint)} \\
 &x_1 + x_2 + x_3 \leq 100 \\
 &\quad \text{(Land constraint)} \\
 &\& x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

The standard form of given LPP is:-
 Introducing slack variables $x_4, x_5, x_6 \geq 0$.

$$\begin{aligned}
 \text{Max } Z &= 100x_1 + 300x_2 + 200x_3 + 0x_4 + 0x_5 + 0x_6 \\
 \text{s.t.} \quad &40x_1 + 20x_2 + 30x_3 + x_4 = 3200 \\
 &x_1 + 2x_2 + x_3 + x_5 = 160 \\
 &x_1 + x_2 + x_3 + x_6 = 100 \\
 &\& x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.
 \end{aligned}$$

Simplex Method:

		$C_j \rightarrow 100 \quad 300 \quad 200 \quad 0 \quad 0 \quad 0$							
Basic Variables	C_B	x_B	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio $x_B/x_k, x_k > 0$
x_4	0	3200	40	20	30	1	0	0	$\frac{3200}{20} = 160$
$\leftarrow x_5$	0	160	1	2	1	0	1	0	$\frac{160}{2} = 80 \leftarrow$
x_6	0	100	1	1	1	0	0	1	$\frac{100}{1} = 100$
		$\Delta_j \rightarrow$	-100	-300	-200	0	0	0	
$R_1' = R_1 - 20R_2$	x_4	0	1600	30	0	20	1	-10	$\frac{1600}{20} = 80$
$R_2' = \frac{R_2}{2}$	x_2	300	80	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{80}{\frac{1}{2}} = 160$
$R_3' = R_3 - 1R_2$	$\leftarrow x_6$	0	20	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{20}{\frac{1}{2}} = 40 \leftarrow$
		$\Delta_j \rightarrow$	50	0	-50	0	150	0	
$R_1' = R_1' - 20R_2'$	x_4	0	800	10	0	0	1	10	-40
$R_2' = R_2' - 20R_3'$	x_2	300	60	0	1	0	0	1	-1
$R_3' = \frac{R_3'}{\frac{1}{2}}$	x_3	200	40	1	0	1	0	-1	2
		$\Delta_j \rightarrow$	100	0	0	0	100	100	

$\therefore$ All $\Delta_j \geq 0$.

$\therefore$ Solution under test is optimal.

Optimal solution :

$$x_1 = 0$$

$$x_2 = 60$$

$$x_3 = 40.$$

$$\begin{aligned} \text{Max } Z &= 100(0) + 300(60) + 200(40) \\ &= 18,000 + 8,000 \\ &= \$26,000 \end{aligned}$$

The farmer has to plant 0 acres of crop A, 60 acres of crop B and 40 acres of crop C to obtain the maximum profit of \$26,000.